Interactive Learning tool for Dyslexia Kids

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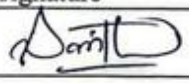
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Dyslexia is a prevalent neurodevelopmental learning difficulty affecting 5–10% of children worldwide, manifesting as persistent difficulties in reading, spelling, and phonological processing. Among the four core skill deficits associated with dyslexia, pronunciation and speech sound processing represent a foundational yet frequently under-addressed area. Inaccurate phoneme production and poor phonological awareness not only impede oral language development but also directly hinder later reading and spelling acquisition. In the Sri Lankan educational context, where Sinhala-medium instruction predominates for young learners, a critical gap exists: no existing digital intervention addresses pronunciation difficulties at the phoneme level in a bilingual (Sinhala-English), adaptive, and culturally contextualized format.

This proposal presents the Pronunciation and Speech Sound Learning Module, an individual component within a larger web-based gamified platform (Project R26-IT-130) targeting dyslexic children aged 5–8 in Sri Lanka. The module employs a listen-repeat-compare interaction model, phoneme-level speech capture via browser-based Web Speech API, and an error-pattern-driven adaptive learning engine to deliver personalized pronunciation practice pathways. Native Sinhala speech samples serve as reference audio, while multimodal feedback mechanisms — including visual waveform comparison and audio playback — provide learners with immediate corrective guidance. An adaptive sequencing algorithm ensures that children receive targeted remediation for their most persistent phoneme-level errors rather than following a fixed, generic lesson order.

The module is built using React.js (frontend), Node.js (backend), and integrates open-source speech processing libraries. Evaluation will employ the System Usability Scale (SUS), pre/post phoneme production accuracy assessments, and qualitative feedback from teachers and speech-language therapists. Expected outcomes include measurable improvements in phoneme production accuracy, enhanced phonological awareness, and reduced severity of common dyslexic speech errors (e.g., phoneme reversals, consonant confusions) in the target population.

Keywords: dyslexia, phonological awareness, speech sound learning, adaptive learning, Sinhala-English bilingual education, gamified intervention, Web Speech API

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My sincere thanks go to the teachers and special education practitioners across Sri Lankan primary schools who were generous with their time and candid about the realities of supporting dyslexic learners in under-resourced classrooms. Their ground-level perspective fundamentally shaped how this module was conceived and what it prioritizes.

To the Dyslexia Association of Sri Lanka thank you for the contextual clarity you provided at a stage in this project when direction was most needed.

Lastly, and most personally, this work is for children who struggle to be heard not because they have nothing to say, but because the sounds do not yet come as easily as the thoughts behind them. It is my hope that this module gives them one more way to practice, one more reason to try, and one more moment of getting it right.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
ASR	Automatic Speech Recognition
COEAL	Centre of Excellence for Artificial Intelligence
IDA	International Dyslexia Association
IT	Information Technology
NLP	Natural Language Processing
SLI	Specific Language Impairment
SLIIT	Sri Lanka Institute of Information Technology
STT	Speech-to-Text
SUS	System Usability Scale
UI	User Interface
UX	User Experience

1. INTRODUCTION

1.1 Background and Context

Dyslexia is recognized internationally as one of the most common neurodevelopmental learning difficulties, estimated to affect between 5% and 10% of school-aged children globally [1]. Characterized by persistent challenges in accurate and fluent word recognition, phonological processing, and spelling, dyslexia presents particular hardships during the early primary years — a period critical for the foundational acquisition of literacy skills. Among the core skill domains implicated in dyslexic profiles, phonological processing, and specifically pronunciation and speech sound production, occupies a central role. Phonological awareness — the ability to recognize and manipulate the sound structure of language — is widely regarded as the strongest predictor of reading success and one of the most reliable early indicators of dyslexic risk [2].

Speech sound difficulties in dyslexic children extend beyond mere articulation they reflect a deeper impairment in phonemic representation and processing. Young dyslexic learners frequently exhibit phoneme confusion (e.g., reversals between /b/ and /d/, /p/ and /q/), inconsistent phoneme-grapheme mappings, and difficulties in segmenting and blending sounds at the sub lexical level. These difficulties, if not addressed through structured and explicit phonological intervention in the early primary years (ages 5–8), carry a significant risk of compounding reading and spelling deficits over time [1][3].

In Sri Lanka, the challenge is compounded by the bilingual nature of early education, where children are expected to develop literacy in both Sinhala (the mother tongue for a majority) and English. According to a 2016 clinical trial registered with the Sri Lanka Clinical Trials Registry, intervention tools for dyslexic children in Sinhala are severely limited, and most available resources are either English-only or lack cultural and linguistic relevance [8]. Additionally, teacher awareness of dyslexia and access to specialized speech-language support in primary schools — particularly in underserved districts — remains critically inadequate [5].

Globally, digital and game-based learning platforms have demonstrated measurable effectiveness in delivering phonological awareness interventions for at-risk young learners. Research by Vasalou et al. (2021) demonstrated that digital game-based phonics instruction significantly improves print knowledge and decoding in pre-readers identified at cognitive risk for dyslexia [6]. However, existing solutions — including internationally recognized platforms such as Lexercise, GraphoGame, and Nessy — are designed primarily for English-speaking learners. None incorporate Sinhala phoneme training, bilingual support, or Sri Lankan curriculum alignment [5][8].

This proposal addresses the gap by presenting an individual research component — the Pronunciation and Speech Sound Learning Module — within a broader collaborative project (R26-IT-130) that aims to develop a comprehensive, web-based gamified platform for dyslexic children aged 5–8 in Sri Lanka. The module specifically targets phoneme-level speech

production accuracy, employing adaptive, error-pattern-driven learning pathways that continuously personalize practice based on each child's unique pronunciation error profile.

This initiative aligns with the United Nations Sustainable Development Goal (SDG) 4 — Quality Education — by promoting inclusive and equitable digital learning tools for children with learning difficulties in a resource-constrained setting. It further aligns with SLIIT's Centre of Excellence for Artificial Intelligence (COEAI) research cluster by leveraging AI-driven adaptive learning and natural language processing technologies to enhance educational outcomes.

1.2 Literature Review

A systematic review of academic literature and existing digital intervention systems reveals important insights into the current state of dyslexia research, phonological intervention methodologies, and available technological tools.

Lyon, Shaywitz, and Shaywitz (2003) established the consensus definition of dyslexia as a specific learning disability characterized by difficulties in accurate and/or fluent word recognition, poor decoding, and poor spelling abilities — difficulties that stem from a deficit in the phonological component of language [1]. This foundational work has guided decades of intervention research and underpins the structured literacy approach endorsed by the International Dyslexia Association [2].

Goswami and Bryant (1990) demonstrated that phonological awareness — encompassing awareness of syllables, onset-rime, and individual phonemes — is a strong causal predictor of reading development [3]. Subsequent research by Bradley and Bryant (1983) confirmed that explicit training in phoneme-level awareness produces significant improvements in reading outcomes, particularly for at-risk learners. These findings directly support the targeted phoneme-level intervention design proposed in this module.

In the digital intervention domain, Segers and Verhoeven (2004) evaluated computer-based phonological training and found significant pre/post improvements in phoneme awareness and reading accuracy among Dutch kindergarten children [4]. Their work provided an early demonstration that structured computer-assisted instruction could replicate and extend the effects of teacher-led phonological training. Similarly, Hatcher et al. (2006) demonstrated that explicit, systematic phonics instruction embedded within engaging practice routines outperformed non-systematic approaches in producing literacy gains for dyslexic learners [9].

Vasalou et al. (2021) conducted a randomized controlled trial in which digital game-based phonics instruction significantly improved print knowledge in pre-readers at cognitive risk for dyslexia compared to a control group, establishing the empirical basis for game-based phonological interventions [6]. The study further highlighted that children engaged more consistently and for longer durations with game-based formats, underscoring the motivational value of interactive design.

Internationally, GraphoGame (Lyytinen et al., 2007) represents the most rigorously evaluated game-based phonological intervention, demonstrating significant improvements in phoneme-grapheme correspondence and reading fluency across multiple languages including Finnish, English, and Swahili [10]. However, GraphoGame does not support Sinhala, and its adaptive algorithms focus on grapheme identification rather than speech production and pronunciation accuracy.

In the Sri Lankan context, the ARUNALU project (SLIIT, 2020) represents a significant attempt to develop a mobile learning ecosystem targeting Sinhala reading weaknesses using voice recognition and natural language processing for phonological training and automated screening [5]. While ARUNALU demonstrates the feasibility of NLP-driven phonological intervention in Sinhala, it focuses on reading comprehension and letter recognition rather than phoneme production at the articulatory level. The LEXIS system (Sabaragamuwa University, 2021) offers web-based screening and basic intervention but lacks adaptive speech production training and bilingual capability [5].

The Sri Lanka Clinical Trials Registry (2023) records a multisensory interactive Sinhala alphabet learning tool evaluated in Gampaha District schools, demonstrating the feasibility of school-based digital interventions for dyslexic children in Sri Lanka [8]. However, the tool focuses on visual and letter-recognition tasks and does not incorporate speech input or phoneme production training.

A review of the literature on feedback mechanisms reveals that immediate, multimodal feedback — combining visual cues and audio playback — is particularly effective for dyslexic learners, who benefit from multisensory reinforcement [2][7]. Ertmer et al. (2007) established that corrective feedback during phonological practice must be specific, timely, and include model-correct-practice cycles to produce durable phonological gains. This informs the listen-repeat-compare interaction model central to this module.

Collectively, the literature establishes a strong empirical foundation for phoneme-level, adaptive, game-based pronunciation intervention. However, no existing system addresses all four of the following dimensions simultaneously: (1) phoneme-level pronunciation analysis rather than word-level evaluation, (2) bilingual Sinhala-English support, (3) adaptive error-pattern-driven personalization, and (4) culturally appropriate content aligned with the Sri Lankan primary curriculum.

1.3 Research Gap

Despite the extensive body of research supporting phonological awareness intervention for dyslexic learners and the availability of several digital tools internationally, a significant and multidimensional research gap exists — particularly in the Sri Lankan educational context.

The most critical gap concerns the granularity of pronunciation assessment. Existing dyslexia tools, including Lexercise, GraphoGame, and even the locally developed ARUNALU, evaluate

pronunciation at the word or syllable level. None perform analysis at the individual phoneme level — the level at which core dyslexic processing deficits originate. This means that a child who produces five phoneme errors within a single word receives a single binary result (incorrect) rather than a diagnostic profile identifying which specific sounds require targeted practice. This blunt assessment approach severely limits the precision of adaptive remediation.

A second major gap concerns bilingual support. Sri Lankan primary children are expected to develop oral language competence in both Sinhala and English. Sinhala phonology differs substantially from English phonology in terms of phoneme inventory, aspiration distinctions, and vowel quality. Existing tools do not address cross-linguistic phoneme interference or provide practice materials in Sinhala that reflect the actual curriculum content encountered by young learners.

Third, no existing locally deployed platform incorporates error-pattern-driven adaptive sequencing — an approach in which the system dynamically constructs each learner's practice pathway based on their individually documented phoneme error history rather than following a fixed, predetermined lesson sequence. This capability is central to delivering the personalized, mastery-based intervention that dyslexic children require.

Fourth, the closed feedback loop between assessment and intervention — where pronunciation assessment results directly and automatically influence subsequent learning task selection — is absent from all reviewed systems. This represents a missed opportunity to operationalize the evidence-based principle of mastery-based progression within a digital platform.

These gaps collectively justify the development and evaluation of the Pronunciation and Speech Sound Learning Module as a novel, evidence-based, and contextually appropriate component of the broader R26-IT-130 platform.

2. OBJECTIVES

2.1 Main Objective

To design, develop, and evaluate a phoneme-level, adaptive Pronunciation and Speech Sound Learning Module within a web-based gamified platform that improves speech sound production accuracy and phonological awareness in dyslexic children aged 5–8 in the Sri Lankan (Sinhala-English bilingual) educational context, through a listen-repeat-compare interaction model supported by error-pattern-driven adaptive practice pathways and multimodal feedback.

2.2 Specific Objectives

1. To develop a browser-based phoneme-level speech capture and analysis pipeline using the Web Speech API that accurately records and segments children's phoneme productions in both Sinhala and English, within a child-friendly interface accessible on standard school devices.
2. To implement a listen-repeat-compare interaction mechanism using native Sinhala and English reference speech samples that exposes dyslexic learners to accurate phoneme models, captures their reproduction attempts, and provides immediate multimodal corrective feedback through visual waveform cues and audio playback comparison.
3. To design and implement an error-pattern detection and profiling system that identifies individual phoneme confusions, substitutions, and mispronunciations at the sub-lexical level, and constructs a dynamic per-learner phoneme error profile over the course of their interaction with the module.
4. To develop an adaptive lesson sequencing algorithm that utilizes the learner's accumulated phoneme error profile to dynamically prioritize and sequence future practice activities, ensuring that the most persistent and unresolved phoneme difficulties receive targeted, spaced, and escalating remediation.
5. To integrate the module with the central student dashboard and analytics engine of the broader R26-IT-130 platform, enabling unified progress tracking, teacher/parent dashboard reporting, and cross-module learning pathway adjustments informed by pronunciation assessment data.
6. To evaluate the module's effectiveness, usability, and educational impact through a pre/post phoneme production accuracy assessment, the System Usability Scale (SUS), and qualitative feedback from teachers and speech-language therapists, targeting a minimum of 80% SUS score and a statistically significant improvement in phoneme accuracy measures.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, which is appropriate for projects that aim to design, construct, and evaluate IT artefacts intended to solve real-world problems. The DSR paradigm, as defined by Hevner et al. (2004), prescribes an iterative cycle of build-evaluate-refine, ensuring that the developed artefact is both technically rigorous and practically relevant. This is supplemented by an experimental pre/post-test design for the evaluation phase to measure the module's educational impact.

3.2 Technical Methods and Algorithms

The core technical approach of the module centers on three interconnected components: speech capture, phoneme analysis, and adaptive sequencing.

Speech Capture and Processing: The module uses the browser-native Web Speech API (Speech Recognition interface) to capture children's spoken phoneme attempts via device microphone. For phoneme-level analysis beyond whole-word recognition, audio input will be pre-processed using the Web Audio API to extract acoustic features relevant to phoneme quality — specifically formant patterns and onset timing. An open-source speech analysis library (such as Praat-via-WebAssembly or CMU Sphinx) will be evaluated for phoneme-level segmentation and feature extraction from the captured audio.

Listen-Repeat-Compare Mechanics: Native speaker recordings of all target phonemes in both Sinhala and English will be produced by a qualified language instructor. For each practice item, the learner listens to the reference audio (played at normal and slow speeds), attempts reproduction via microphone, and receives automated comparison feedback in the form of visual amplitude waveform display and binary correct/incorrect feedback at the phoneme level. Incorrect attempts trigger audio playback of the reference alongside the learner's recording for self-correction.

Adaptive Error-Pattern Sequencing: A weighted error-priority algorithm will be implemented to govern lesson sequencing. Each phoneme in the target phoneme inventory will be assigned a dynamic difficulty weight calculated from: (a) cumulative error frequency across sessions, (b) recency of errors (exponential decay function), and (c) cross-phoneme confusion frequency (e.g., frequency of /b/-/d/ confusion). The practice queue for each session will be generated by sampling phonemes in proportion to their current difficulty weights, ensuring that persistent errors receive frequent, spaced re-exposure consistent with the spaced repetition principles of effective memory consolidation.

3.3 Tools and Platforms

The following tools and platforms are used across the four technical components of the module. The section is organized into: (A) Frontend and UI, (B) Backend and API, (C)

Machine Learning Pipeline, (D) Data Model and Database, and (E) Testing and Quality Assurance.

A — Frontend and User Interface		
Tool / Library	**Role in Module**	**Justification**
React.js	Web frontend framework	Chosen for its component-based architecture, efficient rendering performance suitable for interactive real-time audio UI, and wide industry adoption. Enables modular development of the listen-repeat-compare interface, waveform display, and phoneme stimulus card components.
Tailwind CSS	UI styling framework	Utility-first CSS framework used to build the child-friendly, responsive interface. Eliminates custom CSS overhead and ensures consistent styling across all device form factors — desktop, tablet, and mobile.
Framer Motion	Animation library	Provides smooth, child-appropriate UI animations including phoneme card transitions, waveform rendering, and feedback animations. Improves engagement and usability for the 5–8 year-old target audience.
Web Speech API	Real-time speech capture	Browser-native API for capturing the child's spoken phoneme attempt via device microphone without requiring external plugins or hardware. Provides the raw audio stream input to both the Web Audio API feature extractor and the ML classification pipeline.
Web Audio API	Real-time acoustic feature extraction	Extracts formant features (F1, F2), amplitude envelope, and waveform data directly in the browser for immediate visual feedback display. Complements the server-side ML model by providing low-latency pre-processing without a server round-trip.

B — Backend and API

Tool / Library	Role in Module	Justification
Node.js with Express	Backend server and REST API	Manages session data, learner profiles, phoneme error logs, and communication with the Central Analytics Engine of the R26-IT-130 platform. Stateless API design supports horizontal scalability for concurrent school users.
ONNX Runtime (onnxruntime-node)	ML model inference in Node.js	Loads and runs the exported ONNX phoneme classification model directly within the Node.js backend, enabling real-time inference without a separate Python server dependency. Achieves target response latency of <500ms under standard school broadband conditions.
FastAPI (Python)	Fallback ML model serving API	Lightweight Python REST microservice deployed as a fallback model serving layer if ONNX-in-Node.js faces checkpoint compatibility issues. FastAPI is selected over Flask for its native async support and automatic OpenAPI documentation.
Firebase Storage / AWS S3	Audio file CDN and storage	Stores and serves native speaker WAV/MP3 reference phoneme recordings via a content delivery network. Ensures low-latency audio delivery to school devices across Sri Lanka regardless of geographic location.
Docker	Containerisation of ML service	Containerises the ML model and its serving environment (ONNX Runtime or FastAPI) for consistent, reproducible deployment across development, testing, and production environments.

C — Machine Learning Pipeline (Offline Training)		
Tool / Library	**Role in Module**	**Justification**
Python 3.10+	Core ML programming language	All feature extraction, model training, evaluation, data augmentation, and ONNX export scripts are written in Python. Python

		is used exclusively for the offline training pipeline; the runtime module uses Node.js.
PyTorch	Deep learning framework	Industry-standard framework for model definition, training loop implementation, CTC loss computation, backpropagation, and ONNX export. HuggingFace Transformers is built on PyTorch, making it the natural framework for Wav2Vec2 fine-tuning.
HuggingFace Transformers	Pre-trained model access and fine-tuning	Provides the Wav2Vec2Phoneme pre-trained checkpoint and the Trainer API. The Wav2Vec2CTCTokenizer maps phoneme labels to integer IDs for CTC training. The Trainer API manages the full training lifecycle — batching, gradient accumulation, learning rate scheduling, and checkpoint saving.
librosa	Audio feature extraction	Extracts MFCC (40 coefficients), Delta/Delta-Delta features, and Mel Spectrograms from raw WAV recordings. Standard Python library for audio and music analysis; used in both the offline training pipeline and the server-side inference preprocessing step.
audiomentations	Training data augmentation	Applies real-time augmentation during training: time-stretching, pitch-shifting, gain adjustment, and background noise injection. Artificially expands the limited Sinhala child speech training dataset and reduces overfitting risk.
scikit-learn	Model evaluation and metrics	Computes per-phoneme accuracy, weighted F1-score, k-fold cross-validation, and the phoneme confusion matrix. The confusion matrix output is exported and used to initialise confusion frequency weights in the Adaptive Sequencing Algorithm.
jiwer	Phoneme Error Rate (PER)	Calculates the primary evaluation metric — Phoneme Error Rate — by counting substitutions, insertions, and deletions

		between predicted and ground-truth phoneme sequences. Target PER: ≤15% on the held-out test set.
ONNX (torch.onnx.export)	Model format conversion	Exports the trained PyTorch Wav2Vec2Phoneme model to the ONNX (Open Neural Network Exchange) portable format, enabling framework-agnostic deployment in the Node.js backend without a Python runtime dependency.
Google Colab / Kaggle GPU	Training compute environment	Cloud GPU environments (T4/A100/P100) used for running the transformer fine-tuning pipeline. Selected because standard laptop hardware lacks sufficient GPU memory for transformer model training.
Audacity	Audio recording and preprocessing	Open-source audio editor used to record, trim, normalise, and clean native Sinhala-English phoneme WAV files before feature extraction. Enables segment-level labelling of the reference audio dataset.

D — Data Model

The module employs MongoDB as its primary data store, chosen for its document-oriented schema which accommodates the flexible, hierarchical structure of per-learner phoneme error profiles. The data model comprises the following four core collections:

Collection	Key Fields	Purpose
learnerProfiles	learnerId, age, grade, language preference, sessionCount, lastActive	Stores anonymised learner identity and demographic data. Serves as the root document to which phoneme error profiles and session records are linked. All identifiable fields are pseudonymised at ingestion using a UUID-based mapping.
phonemeErrorProfiles	learnerId, phonemeId, language, errorFrequency, lastErrorTimestamp,	The central data structure of the module. One document per learner per phoneme, updated after every session attempt. Stores the cumulative error frequency,

	confusionMap { targetPhoneme: count }, difficultyWeight	timestamp of the most recent error (for recency decay computation), and a confusion map recording which incorrect phonemes the learner substituted for the target. The difficultyWeight field is recomputed by the Adaptive Sequencing Algorithm at session start and stored for audit purposes.
sessionLogs	sessionId, learnerId, timestamp, duration, practiceQueue [{ phonemeId, attempt, mlClassification, isCorrect, confusedWith }], accuracyRate	Records the complete interaction log for each session. Each entry in the practiceQueue array captures the target phoneme, the ML model's classification result, the binary correctness outcome, and the phoneme the learner was confused with (if incorrect). This log is the primary input for updating phonemeErrorProfiles after each session and for generating performance summaries sent to the Central Analytics Engine.
referenceAudio	phonemeId, language, difficultyLevel, curriculumGrade, audioUrl (CDN), speakerType, duration	Stores metadata for all native speaker reference phoneme recordings. The audioUrl field points to the CDN-hosted WAV/MP3 file on Firebase Storage or AWS S3. Documents are tagged with language (Sinhala/English), curriculum grade alignment (Grade 1–3), and difficulty level to support graduated phoneme introduction in early sessions.

E — Testing and Quality Assurance		
Tool / Library	**Role in Module**	**Justification**

Jest	Unit testing — frontend and backend	Tests individual React components and Node.js API endpoints in isolation. Ensures that feature extraction functions, error profile update logic, and adaptive queue generation produce correct outputs across edge cases.
React Testing Library	Component integration testing	Tests the listen-repeat-compare UI components including audio playback, waveform rendering, and feedback display under simulated phoneme attempt scenarios.
Postman / REST Client	API endpoint testing	Used to validate Node.js REST API endpoints during development, including the phoneme classification inference endpoint, session log POST, and error profile GET/PUT operations.
Manual usability testing	Child-facing interface evaluation	Conducted with target school children in controlled classroom settings with teacher oversight. Evaluates navigation intuitiveness, audio clarity, feedback comprehension, and engagement sustain. Informs iterative UI improvements before the formal SUS evaluation phase.

Summary of All Tools and Platforms

The following table provides a consolidated reference of all tools and platforms used across all five sections of the module:

#	Tool / Library	Category	Role in Module
1	**React.js**	Frontend	Component-based interactive child-facing UI — phoneme cards, waveform display, audio feedback
2	**Tailwind CSS**	Frontend	Responsive, accessible styling across all device types
3	**Framer Motion**	Frontend	Child-friendly UI animations and waveform transitions
4	**Web Speech API**	Frontend	Browser-native real-time phoneme attempt capture via microphone

5	**Web Audio API**	Frontend	Real-time acoustic feature extraction and waveform rendering
6	**Node.js / Express**	Backend	REST API for session management, error profile updates, analytics push
7	**ONNX Runtime (Node.js)**	Backend	Runs phoneme classification model inference in Node.js — no Python server needed
8	**FastAPI (Python)**	Backend	Fallback ML model serving microservice if ONNX-in-Node.js not used
9	**Firebase Storage / AWS S3**	Backend	CDN-hosted native speaker reference audio WAV/MP3 files
10	**Docker**	Backend	Containerises ML model service for consistent deployment
11	**Python 3.10+**	ML Pipeline	Core language for all offline training, evaluation, and export scripts
12	**PyTorch**	ML Pipeline	Deep learning framework for Wav2Vec2 fine-tuning and ONNX export
13	**HuggingFace Transformers**	ML Pipeline	Wav2Vec2Phoneme pre-trained checkpoint and Trainer API for fine-tuning
14	**librosa**	ML Pipeline	MFCC, Mel Spectrogram, Delta feature extraction from audio recordings
15	**audiomentations**	ML Pipeline	Training data augmentation — time-stretch, pitch-shift, noise injection
16	**scikit-learn**	ML Pipeline	Confusion matrix, F1-score, k-fold cross-validation evaluation
17	**jiwer**	ML Pipeline	Phoneme Error Rate (PER) computation — primary model metric
18	**ONNX (torch.onnx.export)**	ML Pipeline	Exports trained model to portable format for Node.js deployment
19	**Google Colab / Kaggle GPU**	ML Pipeline	Cloud GPU environment for transformer fine-tuning

20	**Audacity**	ML Pipeline	Records and preprocesses native Sinhala-English phoneme reference audio
21	**MongoDB**	Data Model	Document-oriented DB — learnerProfiles, phonemeErrorProfiles, sessionLogs, referenceAudio collections
22	**Jest**	Testing	Unit testing — React components and Node.js API endpoints
23	**React Testing Library**	Testing	Component integration testing — UI interaction simulation
24	**Postman / REST Client**	Testing	API endpoint validation during development
25	**Manual usability testing**	Testing	Child-facing interface evaluation with target school children

3.4 Data Collection and Ethical Considerations

Primary data will be collected from two sources: (1) system-generated interaction data (phoneme attempt recordings, error logs, session durations, accuracy rates, progression timestamps) automatically captured by the module during all user sessions; and (2) structured teacher/parent feedback collected through validated questionnaires at the end of the evaluation phase.

Pre/post assessments of phoneme production accuracy will be administered using a standardized phoneme elicitation task battery adapted from the Goldman-Fristoe Test of Articulation-3 (GFTA-3) protocol, contextualized with Sinhala phoneme targets. Assessments will be conducted by trained research assistants in school settings.

All data collection activities will be conducted only after obtaining ethical approval from the SLIIT Research Ethics Committee, written informed consent from parents or legal guardians, child assent, and official written permission from school principals and the relevant provincial education ministry. Participant data will be fully anonymized, stored in password-protected encrypted cloud storage, and accessible only to the research team. Audio recordings of children will be used solely for phoneme analysis within the research context and will not be retained beyond the study period. All ethical procedures will comply with the SLIIT Research Ethics Framework 2024.

3.5 Validation Strategy and Performance Metrics

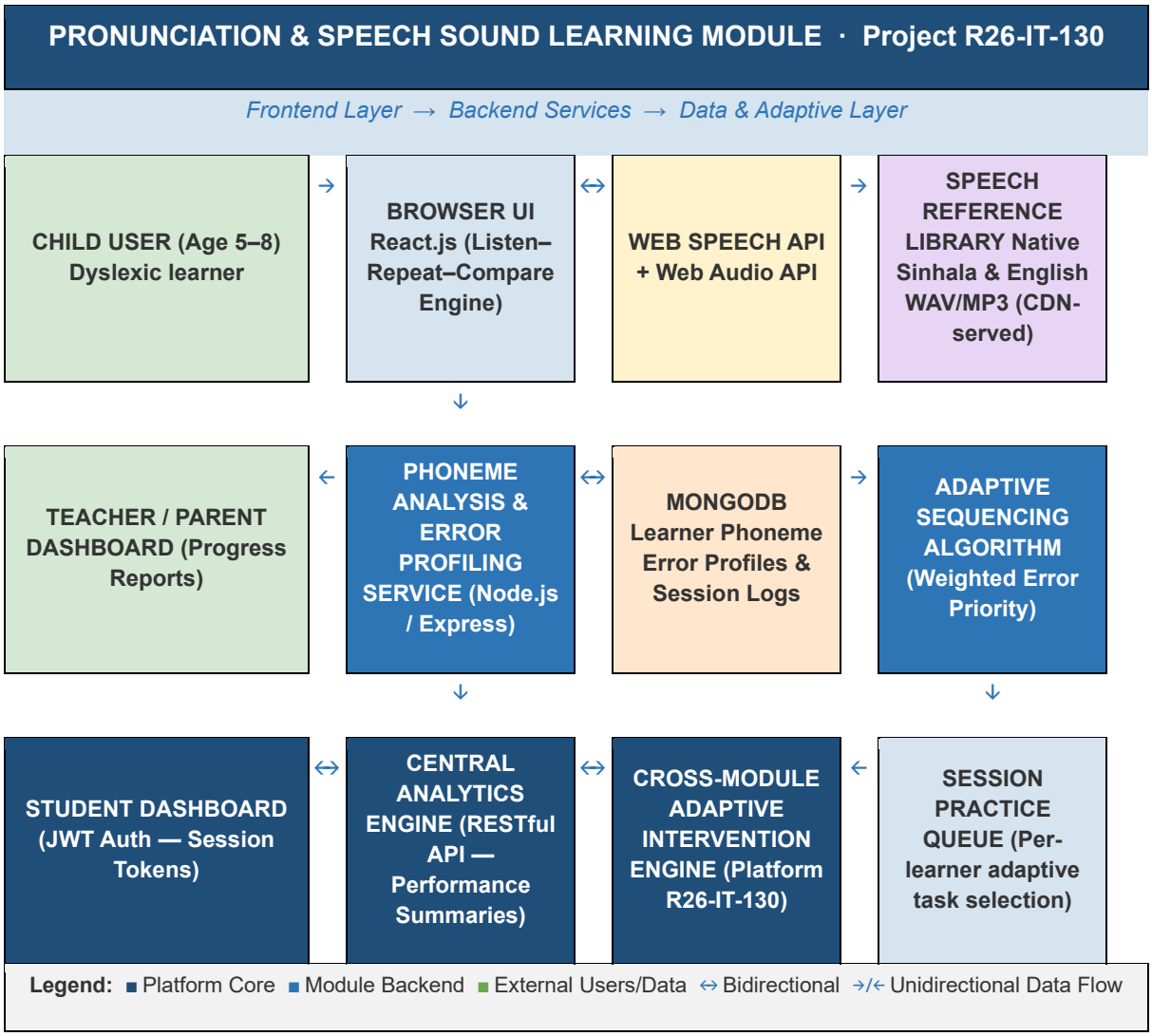
The module will be evaluated across three dimensions:

- Educational Effectiveness: Pre/post comparison of phoneme production accuracy scores for target phonemes, analyzed using paired-samples t-tests. A minimum improvement of 15 percentage points in phoneme accuracy is targeted.
- Usability: System Usability Scale (SUS) administered to teachers and parents post-evaluation. A minimum SUS score of 80 (categorized as 'Good') is targeted.
- Engagement: Session duration, task completion rates, and return visit frequency will be recorded as proxy indicators of learner motivation and platform engagement.
- Technical Performance: Phoneme capture accuracy, response latency (<500ms), and cross-browser compatibility will be assessed through systematic technical testing.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The Pronunciation and Speech Sound Learning Module is one of four skill-specific components within the broader R26-IT-130 gamified learning platform. It operates as a semi-independent microservice module within the platform's architecture, communicating with the Central Student Dashboard, the Analytics Engine, and the Adaptive Intervention Engine through RESTful API calls.

Figure 1: High-Level System Architecture — Pronunciation & Speech Sound Learning Module (R26-IT-130)



4.1 Component Overview

The module comprises four primary internal sub-components:

- Speech Reference Library: A curated repository of professionally recorded native Sinhala and English phoneme audio samples, stored in cloud object storage and served on demand via a CDN. Files are encoded as WAV/MP3 and annotated with phoneme labels, language, difficulty level, and curriculum alignment tags.
- Listen-Repeat-Compare Engine: The interactive UI layer. The child is presented with a phoneme stimulus card, listens to the reference audio, records their attempt via

microphone (Web Speech API / Web Audio API), and receives immediate visual and audio feedback. The engine handles recording buffering, playback synchronization, and comparative waveform rendering.

• Phoneme Analysis and Error Profiling Service: A backend Node.js service that processes each speech submission, extracts acoustic features (energy envelope, fundamental frequency, duration), performs phoneme-level classification against reference patterns, and logs outcomes to the learner's persistent phoneme error profile in MongoDB.

• Adaptive Sequencing Algorithm: A server-side algorithm that reads the learner's current phoneme error profile at the start of each session and generates a weighted practice queue prioritizing phonemes with the highest error frequency, recency, and confusion rates. The queue is refreshed after each session based on updated performance data.

4.2 Data Flow

The data flow within the module proceeds as follows: the child's interaction (listen → repeat → submit) generates a raw audio submission that is captured by the browser and transmitted to the Phoneme Analysis Service. The service processes and logs the result to the learner profile in MongoDB. At session start and end, the Adaptive Sequencing Algorithm reads and updates the phoneme error weights. Performance summaries are transmitted to the Central Analytics Engine, which updates the Teacher/Parent Dashboard in real time. The Student Dashboard retrieves the next session's practice queue from the Adaptive Sequencing Algorithm at the start of every new session.

4.3 Integration with Platform

The module integrates with the broader R26-IT-130 platform via the following interfaces:

- Authentication: Shared JWT-based session tokens managed by the platform's central authentication service.
- Student Profile API: Read/write access to child age, grade, language preference, and cross-module performance scores.
- Analytics Engine API: Pushes per-session phoneme accuracy summaries and error pattern data to the centralized analytics service.
- Adaptive Intervention Engine: Receives cross-module performance data to inform holistic learning pathway adjustments that span all four skill domains.

4.4 Scalability and Security Considerations

The module is designed for horizontal scalability through stateless API design and cloud-based audio storage. Audio submissions from multiple concurrent users are handled asynchronously to prevent processing bottlenecks. Security measures include HTTPS encryption for all data transmission, child data anonymization at the database level, role-based access control for teacher and parent accounts, and compliance with Sri Lanka's Data Protection Bill (2022) provisions concerning minors' data.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Primary User: Dyslexic children aged 5–8 enrolled in Sri Lankan government and private primary schools.
- Secondary Users: Class teachers and special education coordinators who use the progress dashboard to monitor and guide learners.
- Tertiary Users: Parents and legal guardians who access progress reports and home-use recommendations.
- Domain Experts: Speech-language therapists and educational psychologists who validate module design and interpret assessment results.
- System Administrator: IT administrators at the school or platform level responsible for account management.

5.2 Functional Requirements

- The system shall allow a child to listen to a reference audio sample of a target phoneme in Sinhala and/or English at normal and reduced speed.
- The system shall capture the child's spoken phoneme attempt via device microphone using the Web Speech API.
- The system shall perform phoneme-level analysis on the submitted audio and classify it as correct or incorrect relative to the reference phoneme pattern.
- The system shall display immediate visual feedback (waveform comparison) and play back the reference audio alongside the child's recording on incorrect attempts.
- The system shall maintain a persistent per-learner phoneme error profile, recording frequency, recency, and type of phoneme errors across all sessions.
- The system shall generate an adaptive practice session queue at the start of each session, prioritizing phonemes based on the learner's current error profile weights.
- The system shall support at minimum 30 target phonemes in Sinhala and 26 target phonemes in English, aligned with the Sri Lankan Grade 1-3 curriculum.
- The system shall provide syllable segmentation and gradual sound combination exercises to scaffold pronunciation from isolated phonemes to consonant clusters and syllables.
- The system shall transmit per-session performance summaries to the centralized Analytics Engine for integration with the Teacher/Parent Dashboard.
- The system shall allow teachers to review individual learner phoneme error profiles and session histories from the Teacher Dashboard.

5.3 Non-Functional Requirements

- Performance: Speech capture-to-feedback latency shall not exceed 500 milliseconds under standard school broadband conditions (5 Mbps).
- Usability: The child-facing interface shall require no reading ability to navigate; all instructions shall be delivered via audio and visual icons only.
- Accessibility: The module shall be accessible on standard school desktop computers, tablets, and mobile devices without additional hardware or software installation.
- Reliability: The system shall maintain 99% uptime during school hours (7:00 AM – 4:00 PM, Monday to Friday).

- Security: All user data shall be encrypted in transit (TLS 1.2+) and at rest (AES-256). Audio recordings shall not be retained beyond the session window without explicit parental consent.
- Compatibility: The module shall function correctly on Google Chrome (v110+) and Mozilla Firefox (v100+) on Windows 10, macOS 12+, Android 10+, and iOS 14+.
- Scalability: The backend architecture shall support at minimum 500 concurrent users without performance degradation.
- Localization: All text displayed in the interface shall support Unicode rendering for Sinhala script (Unicode block: U+0D80 to U+0DFF).

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market encompasses government and private primary schools in Sri Lanka seeking digital tools to support learners with dyslexia or reading difficulties. Sri Lanka has approximately 9,600 government schools with primary sections, collectively enrolling over 1.5 million primary-level pupils annually. Given the estimated 5–10% dyslexia prevalence, the potential addressable market exceeds 75,000–150,000 dyslexic learners in the primary school system alone.

Secondary markets include private special education centers, child development clinics, and speech-language therapy practices that serve dyslexic children, as well as individual parents seeking supplementary digital home learning resources.

6.2 Value Proposition

The module offers a uniquely contextualized, bilingual (Sinhala-English), phoneme-level adaptive pronunciation learning experience — a capability absent from all currently available local and international tools. Its web-based, device-agnostic design eliminates the need for expensive hardware or software procurement, making it practically deployable in resource-constrained Sri Lankan school environments.

6.3 Revenue Model

- Institutional Subscription: Annual licensing fee per school (proposed: LKR 25,000–50,000 per annum depending on student enrolment). Includes access for all enrolled students, teacher dashboard, and progress reporting.
- Freemium Access: A free-tier module offering a limited phoneme practice set to facilitate adoption by under-resourced government schools, with premium adaptive features available under subscription.
- Ministry Partnership: Partnership model with the Sri Lanka Ministry of Education for subsidized national-level deployment, funded through existing digital learning budget allocations.
- Clinical Licensing: Specialized licensing for speech-language therapy clinics, including extended assessment reports and therapist annotation features.

6.4 Competitive Advantage

The module's competitive advantages include: (1) Sinhala phoneme support — no competitor offers this; (2) phoneme-level (sub-word) analysis granularity; (3) error-pattern adaptive sequencing; (4) Sri Lankan curriculum alignment; and (5) zero additional hardware requirement. These advantages collectively create a defensible niche with high switching costs.

6.5 Intellectual Property Considerations

The native Sinhala phoneme audio recordings will constitute original intellectual property of the research team and will be licensed under a Creative Commons Attribution-Noncommercial license for educational use. The adaptive sequencing algorithm and phoneme error profiling methodology will be documented in a technical report submitted for IP review under SLIIT's Innovation and IP Policy.

7. BUDGET AND JUSTIFICATION

The following budget estimate covers the direct costs associated with developing, deploying, and evaluating the Pronunciation and Speech Sound Learning Module. All costs are presented in Sri Lankan Rupees (LKR).

Item	Category	Estimated Cost (LKR)	Justification
Web hosting (1 year)	Cloud/Hosting	12,000	Deployment of the web-based platform on a reliable server
Domain name registration	Cloud/Hosting	3,000	Custom domain for the application
Microphone API testing devices	Hardware	5,000	Mobile/tablet devices for testing speech input functionality
Native Sinhala speech recording	Data/Content	10,000	Professional native speaker recordings for phoneme samples
Open-source STT API (quota)	Software/API	8,000	Speech-to-text processing costs for pronunciation analysis
Research literature access	Academic	4,000	IEEE/ACM digital library access for research papers
Usability testing incentives	Testing	5,000	Token gifts for school participants and teachers
Miscellaneous	Other	3,000	Contingency for unexpected costs
TOTAL		50,000	

Note: Development effort (student time) is contributed as academic work and is not included in the direct cost estimate. Open-source tools (React.js, Node.js, MongoDB Community Edition) are used to minimize software licensing expenditure.

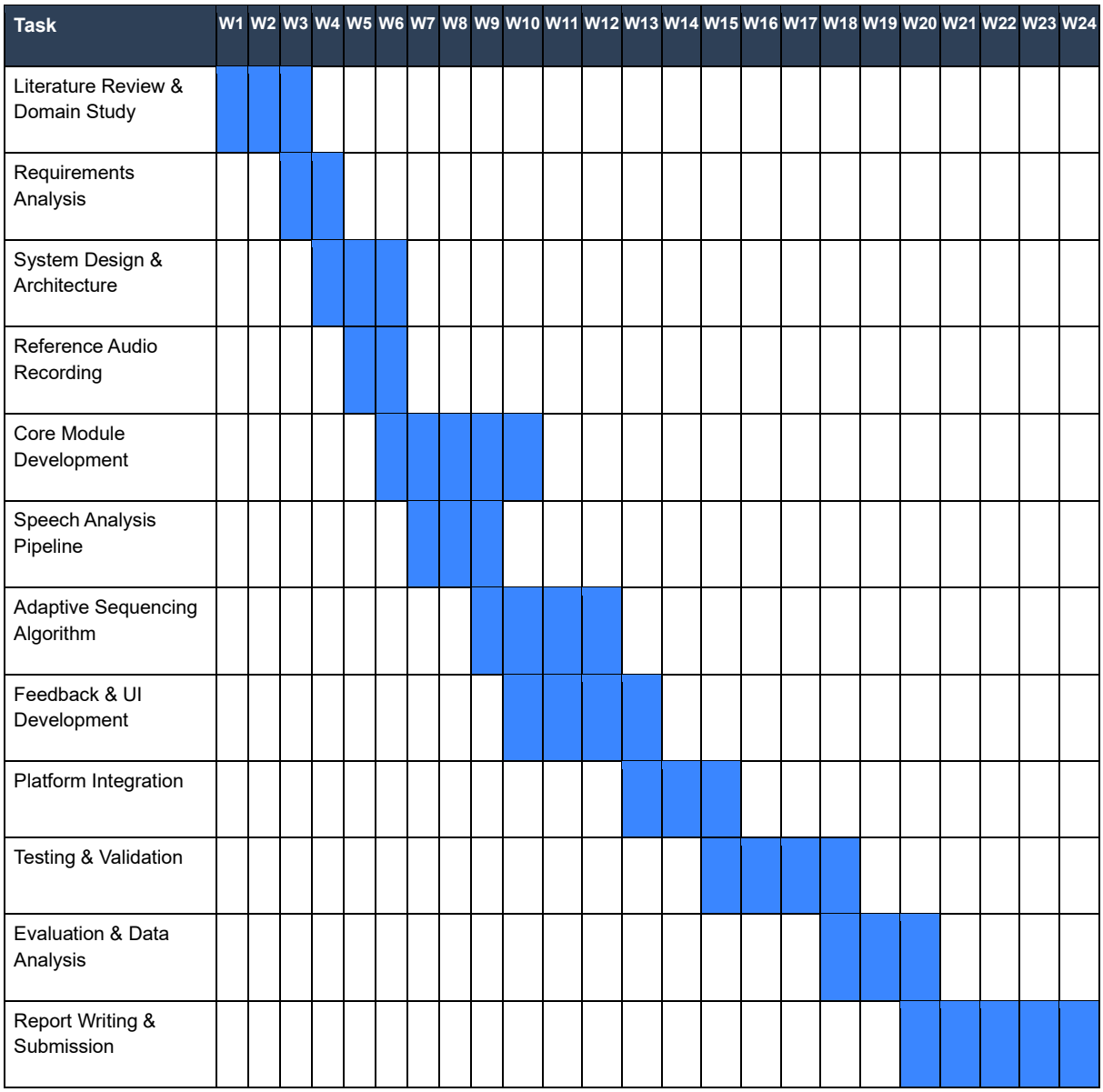
8. WORK BREAKDOWN STRUCTURE (WBS)

The following table presents the Work Breakdown Structure for the Pronunciation and Speech Sound Learning Module, covering all phases from initial research through evaluation and final reporting.

Phase	Task	Sub-Tasks	Duration
1	Literature Review & Domain Study	Review dyslexia, phonological processing, SLI literature; study Sri Lankan context	Weeks 1-3
2	Requirements Analysis	Stakeholder interviews, functional & non-functional requirements definition	Weeks 3-4
3	System Design	Architecture design, UI wireframes, database schema, speech pipeline design	Weeks 4-6
4	Core Module Development	Implement listen-repeat-compare engine, phoneme capture, speech analysis pipeline	Weeks 6-10
5	Feedback & Adaptive Logic	Build visual/audio feedback UI, implement error-pattern adaptive sequencing	Weeks 10-13
6	Integration	Integrate with central dashboard, connect to analytics and progress tracking	Weeks 13-15
7	Testing & Validation	Unit testing, user testing with school children, usability evaluation	Weeks 15-18
8	Evaluation & Report Writing	Data analysis, SUS evaluation, dissertation writing and submission	Weeks 18-22

9. GANTT CHART

The following Gantt chart presents the semester-wise development timeline for the module, spanning Semester 1 (Weeks 1–12) and Semester 2 (Weeks 13–24).



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11. APPENDICES

Appendix A: Phoneme Target List – Sinhala (Preliminary)

The following Sinhala phoneme categories are targeted in the module, aligned with Grade 1–3 Sri Lankan curriculum:

- Vowels: /a/, /aa/, /i/, /ii/, /u/, /uu/, /e/, /ee/, /o/, /oo/
- Stops: /k/, /kh/, /g/, /gh/, /t/, /th/, /d/, /dh/, /p/, /ph/, /b/, /bh/
- Nasals: /m/, /n/, /ng/, /ny/
- Fricatives and approximants: /s/, /sh/, /h/, /w/, /y/, /r/, /l/

Appendix B: Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
Web Speech API accuracy insufficient for phoneme-level analysis	Medium	High	Supplement with Web Audio API acoustic feature extraction; fallback to coarser word-level feedback if phoneme analysis is unreliable.
Insufficient school participation for evaluation	Low	High	Pre-establish MoU with two partnering schools; obtain ethics approval in Semester 1.
Native speaker recording quality issues	Low	Medium	Use professional recording studio; validate recordings with speech-language therapist before integration.
Sinhala Unicode rendering issues across browsers	Medium	Medium	Test on all target browsers in Semester 1; use standard Unicode libraries.
Child disengagement during evaluation sessions	Medium	Medium	Design sessions under 15 minutes; use gamification rewards; involve teachers in session facilitation.

Appendix C: Evaluation Instrument – System Usability Scale (SUS) Items

The following 10 SUS items will be administered to teachers and parents following the evaluation phase, adapted for the educational technology context:

- I think that I would like to use this system frequently.
- I found the system unnecessarily complex.
- I thought the system was easy to use.
- I think that I would need the support of a technical person to be able to use this system.
- I found the various functions in this system were well integrated.
- I thought there was too much inconsistency in this system.
- I would imagine that most people would learn to use this system very quickly.
- I found the system very cumbersome to use.
- I felt very confident using the system.
- I needed to learn a lot of things before I could get going with this system.

Appendix D: Ethical Clearance Checklist

- SLIIT Research Ethics Committee application submitted: [To be completed]
- Informed consent form (parent/guardian) drafted: [To be completed]
- Child assent protocol prepared: [To be completed]
- School principal consent letter template prepared: [To be completed]
- Data anonymization protocol documented: [To be completed]
- Audio recording retention and deletion policy documented: [T0 be completed]